Quantum Interferometry: In the Optical Near and Far-Field

Joseph Nasrollahzadeh • 20561138
School of Physics and Astronomy
University of Nottingham
13th May 2026

Abstract

This report investigates the wave nature of light across both the Fraunhofer (far-field) and Fresnel (near-field) diffraction regimes. Using a $633\ nm$ Helium-Neon (HeNe) laser, far-field single-slit diffraction was used to test the Heisenberg Uncertainty Principle. Analysis yielded a momentum-position uncertainty gradient of $(1.82 \pm 0.17) \times 10^{-33}$ kg m^{2}s^{-1}, equating to approximately $32\ \hbar/2$. This expected inflation arises from defining uncertainty strictly by the physical slit width and primary diffraction nulls, which overestimates the spatial bound when compared to the Gaussian standard-deviation convention. In the near-field regime, the spatial evolution of the interference pattern from a $5\ lines\ /\ mm$ diffraction grating was swept via longitudinal translation of a CCD camera to construct a high-resolution 'Talbot carpet'. Fractional self-imaging was successfully observed at $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ of the Talbot distance. By accounting for a mechanical offset at the sensor plane, the fundamental Talbot distance was measured as $Z_T\ = 12.6\ cm$, showing precise agreement with theoretical predictions.

Table of Contents

Abstract 1
1. Introduction 2
   1.1 Diffraction and Quantum Emergence 2
   1.2 Near-Field Interference and the Talbot Effect 2
   1.3 Project Aims 3
2. Theory 3
   2.1 Diffraction and Propagation Regimes 3
   2.2 Fraunhofer Diffraction and the Heisenberg Uncertainty Principle 4
   2.3 Talbot Distance and Quantum Carpets 5
3. Methodology 6
   3.1 Apparatus and Optical Bench Configuration 6

3.2 Fraunhofer Single-Slit Diffraction 6
3.3 Fraunhofer Multi-Slit Interference 7
3.4 Fresnel Near-Field and Talbot Imaging 8
4. Results 10
4.1 Far-Field Single-Slit Diffraction 10
4.2 Far-Field Grating Interference 11
4.3 Near-Field Talbot Effect 12
5. Discussion 13
5.1 Far-Field Single-Slit and the Uncertainty Principle 13
5.2 Amplitude Modulation in Far-Field Grating Interference 14
5.3 Near-Field Talbot Carpet and Systematic Geometric Correction 14
6. Conclusion 15
7. References 15
8. Appendices 16

1. Introduction

1.1 Diffraction and Quantum Emergence

The propagation of light through confining apertures produces diffraction, a fundamental phenomenon that illustrates the wave nature of light and matter. While the interaction of electromagnetic waves with periodic structures can be entirely described by the Huygens-Fresnel principle, the mathematical framework governing the propagation is notoriously complex [1]. By observing light across different propagation distances, it is possible to find macroscopic emergence of quantum mechanical phenomena.

One such phenomenon is the Heisenberg Uncertainty Principle, a fundamental cornerstone of quantum mechanics. This principle dictates that a particle's spatial position and momentum are complementary variables. This relationship has been rigorously verified not only for photons but also for massive, complex structures such as C_{70} fullerene molecules at high temperatures [2]. By observing the far-field propagation of a wave diffracted through a single slit, the fundamental nature of quantum complementarity can be visualised on a macroscopic scale.

1.2 Near-Field Interference and the Talbot Effect

While far-field diffraction simplifies to a continuous momentum spread, near-field diffraction through a periodic structure introduces a complex wave phenomenon. First

observed by H. F. Talbot in 1836, the Talbot effect represents an intriguing macroscopic lens-less self-imaging phenomenon [3]. Rather than spreading uniformly, the propagating wavefront undergoes periodic self-imaging alignment.

When observed continuously along the propagation axis, these repeating self-images form a highly structured interference pattern known as Talbot carpets, or "quantum carpets". Having a highly coherent laser source is essential to create these carpets as phase inconsistency would degrade the interference patterns [4].

1.3 Project Aims

This project aims to experimentally characterise the near-field and far-field interference patterns of a $633\ nm$ Helium-Neon (HeNe) laser, bridging the classical transition between the Fresnel and Fraunhofer optical regimes. To achieve this, the study focuses on these primary objectives: investigating single-slit diffraction to find the momentum-position coefficient, thus testing the limits of the Heisenberg Uncertainty Principle; Evaluating the far-field distribution generated by a multi-slit diffraction grating; and exploring near-field interference by sweep-scanning the longitudinal optical axis to observe the Talbot effect and identify self-imaging nodes in order to construct a high-resolution Talbot carpet.

2. Theory

2.1 Diffraction and Propagation Regimes

To model the intensity distribution at an observation plane, the physical distance between the aperture and the plane leads the mathematical approximations. When the observation plane is located at a short distance from the aperture, the curvature of the propagating wavefronts is significant. This is known as the Fresnel, or near-field, diffraction regime [5]. The condition for Fresnel diffraction is formally defined as,

$$L \leq \frac{a^2}{\lambda}, \qquad \{1\}$$

where L is the propagation distance from the aperture to the observation plane, a is the characteristic width of the aperture, and λ is the wavelength of the incident light.

Conversely, when the light is allowed to propagate over a sufficiently large distance, the wavefronts expand to be mathematically treated as planar. This far-field limit is defined by the Fraunhofer condition [5], requiring that,

$$L \gg \frac{a^2}{\lambda}, \qquad \{2\}$$

Fraunhofer diffraction is demonstrated in Figure 1.

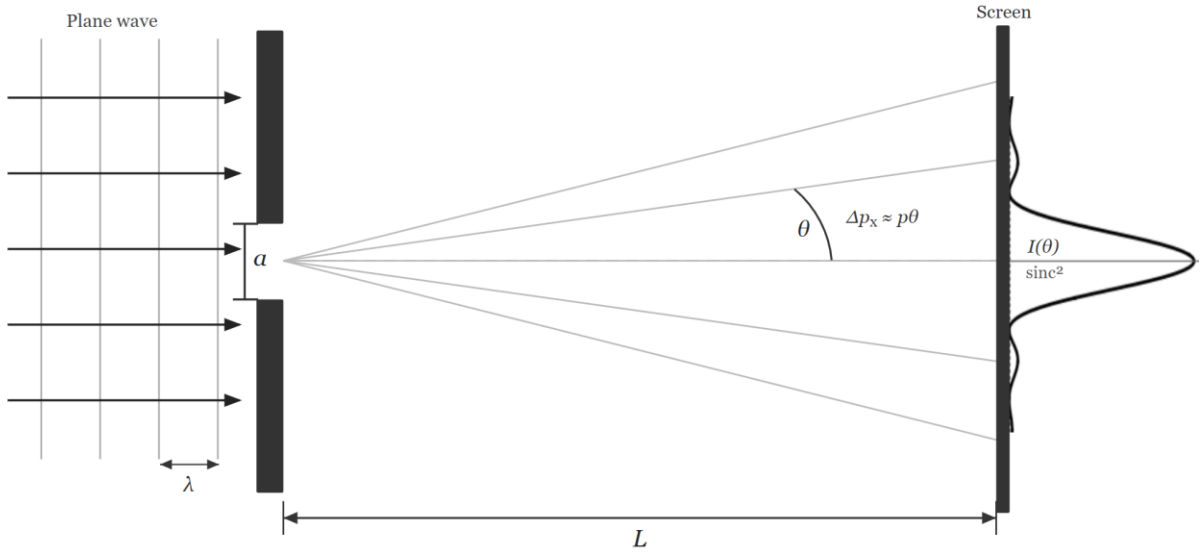


Figure 1: Geometric schematic of single-slit Fraunhofer diffraction. A monochromatic plane wave of wavelength λ is incident on a slit of width a. The diffracted wavefront propagates a distance L to an observation screen, where the characteristic sinc² intensity envelope is formed. The angle θ to the first diffraction minimum defines the transverse momentum spread $\Delta p_x \approx p\theta$. The far-field (Fraunhofer) condition $L \gg a^2/\lambda$ ensures wavefronts at the observation plane are approximated as planar.

Under the Fraunhofer condition, the quadratic phase terms in the diffraction integral become negligible, and the entire expression simplifies. The resulting Fraunhofer diffraction pattern is simply the Fourier transform of the aperture's transmission function, making it advantageous for both analytical and computational modelling [6].

2.2 Fraunhofer Diffraction and the Heisenberg Uncertainty Principle

In the Fraunhofer regime, the intensity distribution of light passing through a uniform single-slit is governed by the squared sinc function,

$$I(\theta) = I_0 \,\mathrm{sinc}^2\left(\frac{\pi a \sin\theta}{\lambda}\right) \qquad \{3\}$$

where I_0 is the peak central intensity and θ is the angular deviation from the optical axis. The uncertainty principle dictates that the product of the uncertainties in a particle's transverse position, Δx , and transverse momentum, Δp_x , must satisfy a fundamental lower bound [7]:

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \qquad \{4\}$$

As a coherent beam of photons (or matter waves) approaches a single slit, its position is relatively unrestricted. However, upon passing through the boundaries of the slit, the transverse spatial uncertainty of the wave is forcefully constrained to the slit width, setting $\Delta x = a$. To satisfy the uncertainty principle, this sudden restriction in spatial position necessitates a proportional increase in momentum uncertainty, Δp_x.

This momentum spread appears physically as the angular divergence of the diffracted beam. If the forward momentum of the wave is $p = h/\lambda$, the transverse momentum required to deflect the wave to the first diffraction minimum (at a small angle $\theta \approx \lambda/a$) can be quantified as $\Delta p_x \approx p\theta$. Substituting in these parameters attains the expected theoretical relationship $\Delta x \Delta p_x = h$.

2.3 Talbot Distance and Quantum Carpets

When a transmission grating of spatial period d is illuminated by a monochromatic plane wave, the diffracted wavelets recombine to produce exact, in-focus images of the original grating at regular longitudinal distances. These self-images occur at integer multiples of the Talbot length [3], Z_T, which is defined as:

$$Z_T = \frac{2d^2}{\lambda} \qquad \{5\}$$

At specific fractions of this characteristic distance, the wavefront exhibits further periodic reconstruction. At half-integer multiples of the Talbot length ($Z_T/2$), an exact image of the grating reappears, but it is shifted laterally by a half-period, $d/2$, creating an "anti-shadow" [3]. The geometric regime of the near-field is revealed at other rational fractions of Z_T, such as $\frac{Z_T}{4}$ or $\frac{3Z_T}{4}$, where the initial phase modulation transitions into amplitude modulation, creating sub-images of a higher frequency and of varying contrasts between the rational fractions.

When the wave intensity is mapped as a continuous function of the longitudinal distance behind the grating, the overlaying integer and fractional self-images come together to form a densely structured, fractal interference pattern. These visually striking, non-synchronous overlapping fields are referred to as Talbot carpets, or "quantum carpets".

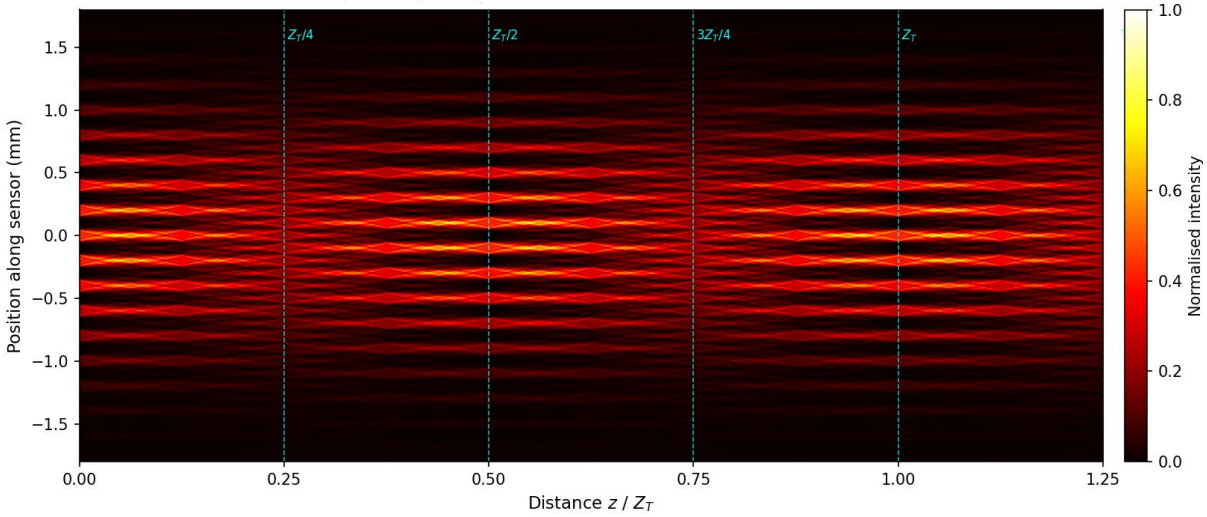


Figure 2. Computational simulation of a near-field Talbot carpet for a periodic diffraction grating of 5 lines/mm, plotting normalised intensity of the beam over the range of the Talbot distance. A

wavelength of 632nm, and beam width of 2mm was computed, resulting in a Talbot distance of 126mm. Self-images of the grating occur at fractional multiples of the Talbot length, $Z_T = \frac{2d^2}{\lambda}$. *The dotted blue bars represent quartiles of the Talbot length, where inverse self-imaging is seen in pairs.*

High source coherence is essential to map these carpets, as an incoherent light source causes the detailed interference patterns to fade due to a low Signal to Noise Ratio.

3. Methodology

3.1 Apparatus and Optical Bench Configuration

The experimental investigations across both the Fraunhofer and Fresnel diffraction regimes were conducted using a continuous Helium-Neon (HeNe) laser source, providing a coherent monochromatic beam at a wavelength of $\lambda = 633\ \mathrm{nm}$. The beam was directed along a precision optical bench, with all components mounted on adjustable optical mounts to ensure strict spatial alignment.

Spatial intensity distributions were recorded using a high-resolution Charge-Coupled Device (CCD) camera, featuring a sensor array of 1280×1024 pixels with a pixel pitch of $s_{\mathrm{pix}} = 5.2 \times 10^{-6}\ \mathrm{m}$. To prevent sensor saturation and ensure that the recorded pixel intensities remained within the linear dynamic range of the 8-bit detector (between 0 and 255), a Neutral Density (ND 1.5) absorptive filter was inserted into the beam path immediately prior to the sensor. This attenuation was critical for accurately resolving the relative amplitudes of the central and higher-order diffraction maxima without clipping the intensity profiles. The filter strength was found experimentally, when the highest peak was found to be in range of the detector.

3.2 Fraunhofer Single-Slit Diffraction

To investigate the far-field single-slit diffraction profile and experimentally test the macroscopic limits of the Heisenberg Uncertainty Principle, an adjustable micrometre slit was introduced into the optical path.

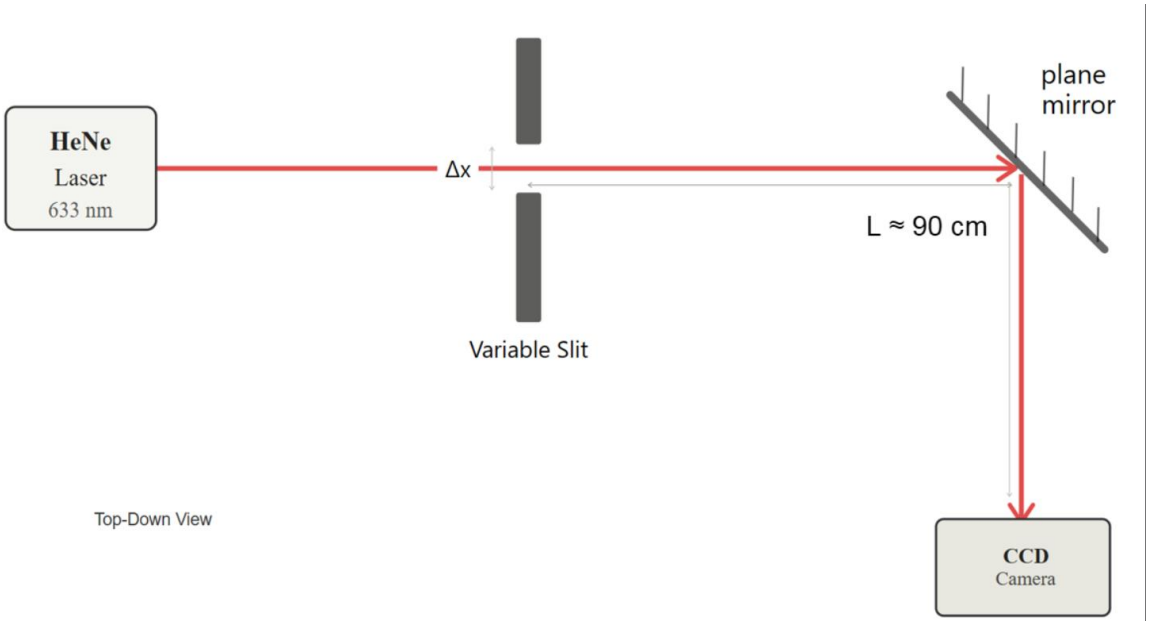


Figure 3. Schematic representation of the Fraunhofer single-slit diffraction optical path. A 633 nm HeNe laser illuminates a variable width slit ($\Delta x = 0.2\ mm$ to $0.8\ mm$). To satisfy the far-field condition ($L \gg \frac{a^2}{\lambda}$), the beam path is redirected via a plane mirror, achieving a total propagation distance of $L \approx 90\ cm$ to the CCD sensor plane. Momentum uncertainty is calculated by measuring the angle to the first diffraction minimum.

The mathematical condition for Fraunhofer diffraction requires the observation distance L to be significantly larger than $\frac{a^2}{\lambda}$. To achieve a sufficient propagation distance within the constraints of the laboratory, a plane mirror was used to turn the optical path, establishing a total slit-to-sensor distance of $L = 0.900 \pm 0.001\ \mathrm{m}$.

The physical slit width, a, was systematically varied from $0.2\ \mathrm{mm}$ to $0.8\ \mathrm{mm}$ in $0.1\ \mathrm{mm}$ increments. For each configuration, a spatial intensity image was captured. The physical width of the slit was taken as the absolute boundary for the transverse positional uncertainty, Δx, of the incident photons.

3.3 Fraunhofer Multi-Slit Interference

To characterise multi-slit interference, the single slit was replaced with a fixed diffraction grating possessing a spatial frequency of $5\ \mathrm{lines/mm}$, and thus a grating period of $d = 0.2\ \mathrm{mm}$.

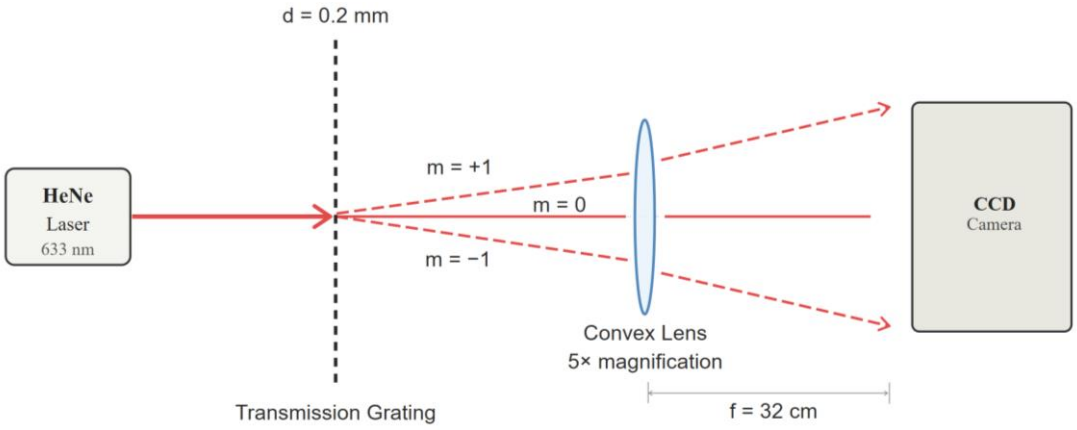


Figure 4. Optical configuration for multi-slit interference using a $5\ lines/mm$ *transmission diffraction grating (*$d = 0.2\ \mathrm{mm}$*). A positive convex lens providing 5x magnification is positioned between the grating and the sensor. This projects the magnified spatial intensity distribution across the sensor array at a fixed lens-to-sensor distance of* $32\ \mathrm{cm}$*, minimising clipping errors across the 8-bit CCD.*

Because the angular separation of the primary interference orders produced by a $0.2\ \mathrm{mm}$ grating is relatively small compared to the CCD sensor, a positive optical convex lens with a $5\times$ magnification factor was introduced between the grating and the camera. This lens magnified the interference pattern, mapping the narrow angular deviations over a broader range of the CCD sensor to reduce resolution error. The distance from the magnification lens to the camera sensor was fixed at $f = 0.32\ \mathrm{m}$, the focal length of the lens.

3.4 Fresnel Near-Field and Talbot Imaging

The final phase of the experiment mapped the near-field interference to construct a Talbot carpet and verify the spatial locations of self-imaging fractions. The $5\ \mathrm{lines/mm}$ diffraction grating was used, and the CCD camera was mounted onto a precision longitudinal translation mount aligned with the vector of propagation, z.

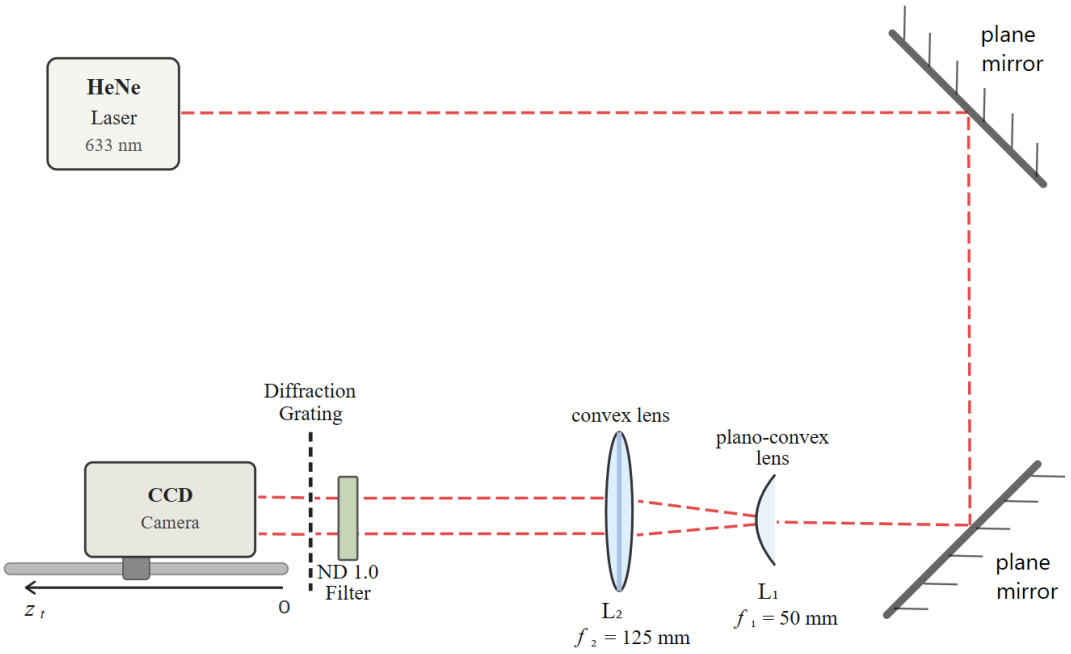


Figure 5. Near-field interferometry setup for mapping the spatial evolution of the Talbot carpet. To ensure an even phase illumination across enough grating slits, the 633 nm HeNe beam is expanded using a Keplerian telescope (50 mm plano-convex and 125 mm convex lenses). The CCD camera is mounted on a longitudinal translational mount to scan the length of axis $\vec{z}$ around the Talbot length of Z_T.

Initial near-field test scans revealed that the narrow diameter of the raw HeNe beam illuminated an insufficient number of grating slits. Because the Talbot effect relies on the transverse superposition of a large array of secondary wavelets, illuminating too few slits truncates this superposition, degrading the fractal self-imaging pattern at longer propagation distances. Initial attempts to expand the beam using a standard telescope and mirror array introduced severe optical misalignment. To resolve this, a dedicated beam-expanding system was constructed using a $125\ \text{mm}$ focal length convex lens and a $50\ \text{mm}$ plano-convex lens, separated by $175\ \text{mm}$. This configuration successfully expanded the beam diameter, ensuring a uniform illumination across the grating surface.

With the expanded beam, a coarse longitudinal sweep was executed by translating the camera from $z = 30\ \text{mm}$ to $z = 150\ \text{mm}$ in increments of $\Delta z = 5\ \text{mm}$. This macroscopic scan identified the regions bounding the primary Talbot distance, Z_T. A subsequent fine scan was performed through the full Talbot distance using increments of $\Delta z = 1\ \text{mm}$. Measurement of z relative to the external housing required a $1.82\ cm$ offset correction due to housing constraints on the camera.

To mitigate the effects of sensor noise and beam misalignment, the CCD and translation mounts were iteratively adjusted and fine-tuned for stable alignment. CCD

exposure parameters and ND filter attenuation were fixed to utilise the full 8-bit range without over-saturation of pixels.

4. Results

4.1 Far-Field Single-Slit Diffraction

Spatial intensity profiles for single-slit widths ranging from $a = 0.2$ mm to 0.8 mm were recorded in the far-field regime ($L = 0.900$ m). Figure 6 displays a selection of these profiles overlaid with their corresponding theoretical sinc^2 fits.

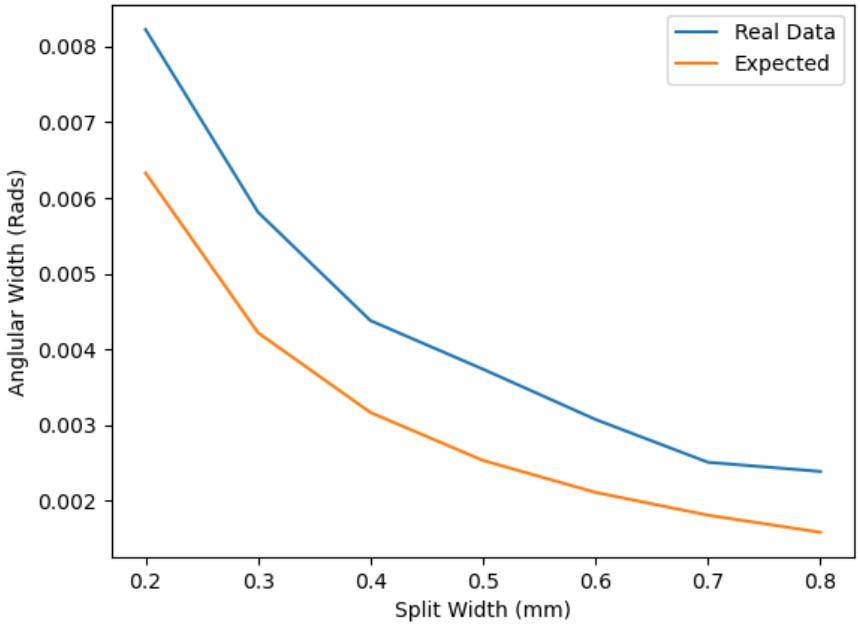


Figure 6: Overlaid far-field spatial intensity profiles for single-slit widths of $a = 0.2$ to 0.8 mm. The raw experimental data is presented alongside the corresponding theoretical sinc^2 distribution fits, computed using eq.3 on given values.

The pixel distance between the primary diffraction minima was extracted for each slit width to calculate the transverse momentum uncertainty, Δp_x. Figure 7 presents the linear relationship between Δp_x and the reciprocal of the transverse positional uncertainty, $1/\Delta x$.

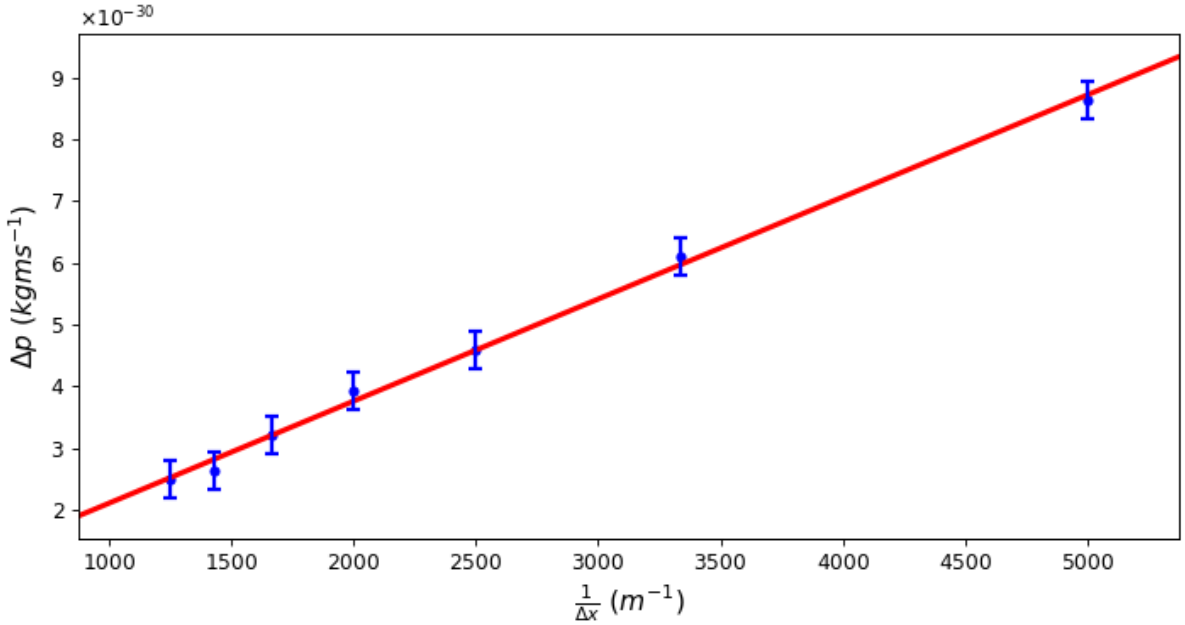


Figure 7. Plot of momentum uncertainty (Δp_x) against the reciprocal of the positional uncertainty ($1/\Delta x$). The error bars represent the propagated uncertainty derived from a conservative ± 50-pixel resolution limit. A linear least-squares fit is applied to the data, yielding a momentum-position uncertainty gradient of $(1.82 \pm 0.17) \times 10^{-33} \mathrm{kg\ m^2 s^{-1}}$.

A linear least-squares fit applied to the data yielded a momentum-position uncertainty gradient of $(1.82 \pm 0.17) \times 10^{-33} \mathrm{kg\ m^2 s^{-1}}$.

4.2 Far-Field Grating Interference

The far-field intensity distribution generated by the $5\ \mathrm{lines/mm}$ diffraction grating ($d = 0.2\ \mathrm{mm}$) is presented in Figure 8. The experimental data is overlaid with the theoretical single-slit sinc^2 envelope to demonstrate the amplitude modulation of the individual interference maxima.

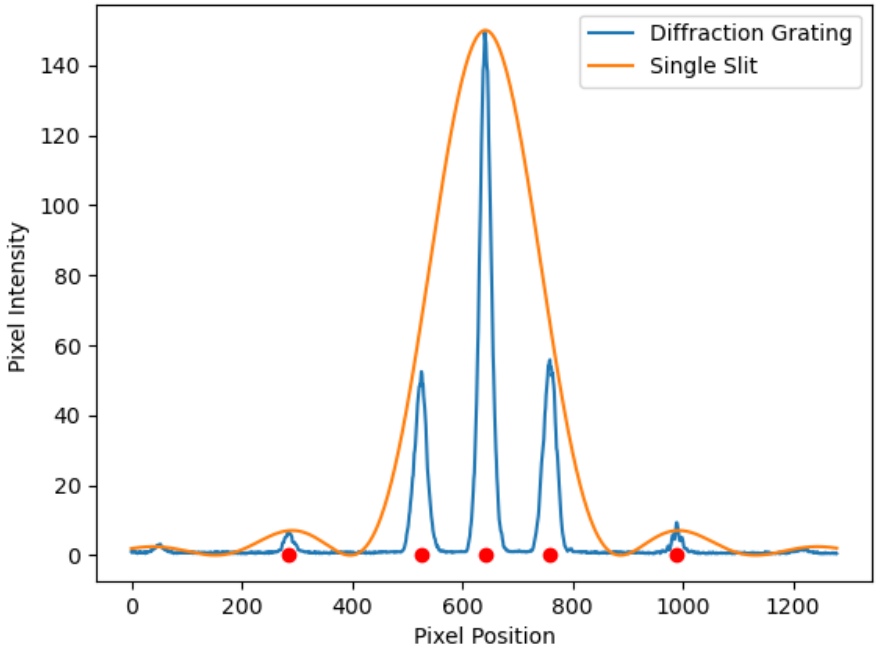


Figure 8: Far-field spatial intensity distribution generated by a $5\ \text{lines/mm}$ *diffraction grating. The raw experimental data (blue) is overlaid with a fitted single-slit* sinc^2 *envelope (orange), demonstrating the amplitude modulation of the higher-order interference maxima. Red markers indicate the central coordinates of the* $m = 0, \pm 1,$ *and* ± 2 *diffraction orders, extracted with the automated Python peak-finding algorithm.*

Table 1 details the measured angular positions of the primary interference peaks alongside the theoretical values calculated using the grating equation, $d \sin\theta = m\lambda$ [6].

Diffraction Order (m)	Measured Angle (rad)	Theoretical Angle (rad)
± 1	± 0.0033	± 0.0032
± 3	± 0.0101	± 0.0095
± 5	± 0.0166	± 0.0158

Table 1. Measured and theoretical angular positions for the $5\ lines/mm$ *diffraction grating interference orders.*

4.3 Near-Field Talbot Effect

Near-field spatial intensity profiles were concatenated to construct a Talbot carpet, mapping the longitudinal progression of the wavefront across the Z axis, as shown in Figure 9.

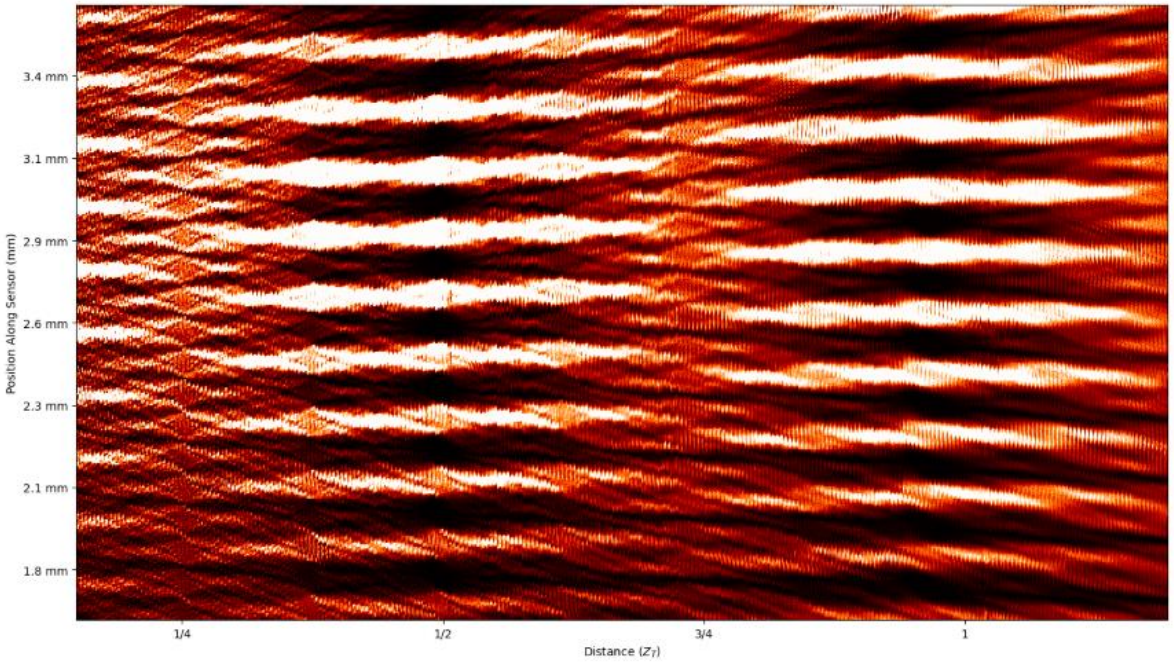


Figure 9. Reconstructed near-field Talbot carpet generated by appending one-dimensional intensity profiles across the longitudinal axis (z). The primary Talbot distance (Z_T) and the fractional self-imaging planes ($\frac{1}{4}Z_T$, $\frac{1}{2}Z_T$, $\frac{3}{4}Z_T$) are visibly resolved within the Talbot carpet.

The locations of the self-images were extracted directly from the array. The longitudinal distances, corrected for a 1.82 cm systematic CCD sensor offset, are as follows:

Fraction of Talbot Length	Longitudinal Distance, z (cm)
$1/4Z_T$	3.15
$1/2Z_T$	6.30
$3/4Z_T$	9.45
Z_T	12.60

Table 2. Talbot length self-imaging fractions in measured longitudinal distance from the diffraction grating to CCD camera sensor.

5. Discussion

5.1 Far-Field Single-Slit and the Uncertainty Principle

The far-field single-slit measurements successfully proved the inverse relationship between a wave's spatial confinement and its subsequent transverse momentum spread. The plotted $\Delta p_x \; vs \; 1/\Delta x$ relationship (Figure 7) yielded a linear trend; providing clear verification of the Heisenberg Uncertainty Principle and satisfying the objective of this investigation.

The linear fit returned a momentum-position uncertainty gradient of $(1.82 \pm 0.17) \times 10^{-33}\mathrm{kg\ m^2 s^{-1}}$. This comfortably satisfies the fundamental Heisenberg Uncertainty lower bound ($\Delta x \Delta p_x \geq \hbar/2 \approx 0.527 \times 10^{-34}\ kg\ m^2\ s^{-1}$) but exceeds the theoretical

minimum by a factor of approximately $32\hbar/2$. This is not an experimental error but a direct consequence of how uncertainty is defined in this measurement.

The $\hbar/2$ minimum applies strictly to ideal Gaussian wave packets, where uncertainties are characterised by their statistical standard deviations (σ). Here, a physical boundary approach was taken instead: Δx was defined as the absolute width of the slit (a), and Δp_x was derived from the total angular spread of the primary diffraction minima ($\Delta\theta \approx 2\lambda/a$). This is a broader, but more rigorous measure than a Gaussian standard deviation, and the inflated coefficient is directly due to that choice.

By contrast, prominent studies, such as Nairz et al. (2001) [2], utilise the FWHM convention, yielding a pre-factor on the order of h ($\approx 12.5\hbar/2$). These larger coefficients are a direct consequence of the physical boundary approach employed, which captures the full extent of the diffracted photons rather than the statistical spread of an ideal Gaussian, and thus naturally inflates the uncertainty.

5.2 Amplitude Modulation in Far-Field Grating Interference

The far-field grating distribution (Figure 8) confirmed a key prediction of multi-slit diffraction theory; that the individual interference maxima are not uniform in intensity but are described by an overlying $sinc^2$ envelope.

This arises because the slits comprising the grating have a finite width and each one independently generates its own Fraunhofer diffraction pattern. The total observed intensity is therefore the product of the grating interference function and the single-slit diffraction envelope, rather than a superposition of point sources. The consequence of this is in the higher-order peaks; wherever a periodic grating maximum coincides with a single-slit minimum, the intensity goes to zero and that diffraction order is negated entirely.

5.3 Near-Field Talbot Carpet and Systematic Geometric Correction

Sweeping the CCD camera in our experimental setup resulted in a successful construction of a Talbot carpet (Figure 9). The repeating self-image structure, observed at the predicted fractional Talbot distances, confirms the lens-less self-imaging documented by H.F. Talbot [3].

The theoretical Talbot distance for a $0.2\ mm$ grating illuminated at $633\ nm$ is $12.64\ cm$ (eq. 5). Initial extraction from the uncorrected longitudinal data placed the primary self-image, and thus the Talbot distance, at $10.78\ cm$, which is 14% below expectation.

This offset occurred because the longitudinal measurement was unable to reach the internal CCD sensor due to the nature of the sensor housing unit. Physical recalibration of the housing revealed a systematic mechanical error of $1.82\ cm$. Applying this correction yields the $12.60\ cm$ primary self-image distance presented in Table 2, agreeing with the theoretical prediction to within 0.4%. The successful discovery and

subsequent solution to this systematic error demonstrates that fine interferometry is critically dependent on rigorous calibration and optical equipment setup.

6. Conclusion

This project successfully achieved its primary objective of bridging the transition between the classical Fraunhofer and Fresnel optical regimes, yielding accurate quantitative verifications of both far-field and near-field wave mechanics.

In the far-field, the spatial spreading of a single-slit diffraction pattern showed a clear macroscopic visualisation of quantum complementarity. Rather than solely affirming classical wave theory, the measured momentum-position uncertainty gradient confirmed that physical apertures enforce the limits of the Heisenberg Uncertainty Principle. Measuring the distance between the diffraction minima further provided visible proof of quantum mechanical emergence from a classical wave. Furthermore, the analysis of multi-slit interference reaffirmed that the interference pattern remains bound by the single-slit diffraction envelope, as predicted from theory.

Conversely, observations in the near-field Fresnel optical regime illustrated the transition from continuous momentum spreading to complex periodic self-organisation. The successful mapping of a high-resolution Talbot carpet confirmed the phenomenon of macroscopic lens-less self-imaging. The challenge of producing accurate and stable interference patterns in the near-field demonstrates that the formation and observation of "quantum carpets" relies heavily on maintaining precise phase coherence and spatial calibration.

Finally, this investigation highlights how simple optical configurations can be used to map profound physical principles, from universal uncertainty limits to fractal wave propagation, and even quantum emergence in macroscopic systems. The future of this field could explore how structured light, such as beams carrying orbital angular momentum, behaves within these interferometric boundaries, potentially discovering new fields in optical information encoding [2].

7. References

[1] M. Born and E. Wolf, *Principles of Optics*, 7th ed. Cambridge, UK: Cambridge University Press, 1999

[2] O. Nairz, B. Brezger, M. Arndt, and A. Zeilinger, "Diffraction of complex molecules by structures made of light," *Phys. Rev. Lett.*, vol. 87, no. 16, p. 160401, Oct. 2001

[3] W. H. F. Talbot, "Facts relating to optical science. No. IV," *Philos. Mag.*, vol. 9, no. 56, Dec. 1836

[4] W. B. Case, M. Tomandl, S. Deachapunya, and M. Arndt, "Realization of optical carpets in the Talbot and Talbot-Lau configurations," *Optics Express*, vol. 17, no. 23, pp. 20966-20974, Nov. 2009

[5] F. L. Pedrotti, L. M. Pedrotti, and L. S. Pedrotti, *Introduction to Optics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2017. pp. 308-333, 267-292

[6] E. Hecht, *Optics*, 5th ed. Harlow, U.K.: Pearson Education, 2017, pp. 520-530

[7] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2018, p. 35

8. Appendices

Automated data processing used a custom Python script to extract, smooth, and isolate interference maxima from one-dimensional pixel intensity profiles, alongside generating two-dimensional Talbot carpets for near-field spatial analysis. The single-slit transverse momentum uncertainty, Δp_y, was determined from the physical separation between primary diffraction minima, Δy . Accounting for physical limits, the combined absolute uncertainty was calculated using the standard partial derivative method:

$$\sigma_{p_y} = \Delta p_y \sqrt{\left(\frac{\sigma_{\text{diff}}}{\Delta y}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2}$$

Code Appendices:

Appendix A: Single-Slit Diffraction Analysis

Loads diffraction images for slit widths 0.2–0.8 mm, detects peak positions using Gaussian smoothing and peak finding, computes the angular width of the central diffraction maximum, and plots measured versus expected angular widths.

```
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import math
from scipy.signal import find_peaks
from scipy.ndimage import gaussian_filter1d

slitRange = np.arange(2, 9, 1)
x = np.arange(0, 1280, 1)
colAvg = 8
meterPerPixel = 5.2e-6  # size of pixel in metres
L = 0.9                  # length from grating to camera
```

```
angleList = []
for slit in slitRange:

im = Image.open(str(math.floor(slit) / 10)      +
'mm.png').convert('L')


    na = np.array(im)

    midRowsAvg = np.average(
        na[((np.shape(na)[0] // 2) - colAvg // 2):((np.shape(na)[0]
// 2) + colAvg // 2), :],
        axis=0
    )
    rowsSmooth = gaussian_filter1d(midRowsAvg, sigma=3)
    fig1 = plt.plot(x, midRowsAvg)

    peaks = find_peaks(rowsSmooth, width=10, distance=150, height=4)

    centralMaxima = np.argmax(rowsSmooth[(peaks[0])])
    if len(peaks[0]) > 2:  # filter out 2nd-order and higher maxima
to get the central two
        difference = (peaks[0][centralMaxima + 1]) -
(peaks[0][centralMaxima - 1])
    elif len(peaks[0]) == 2:  # for images where both outer maxima
couldn't fit; use centre and one side instead
        difference = (peaks[0][1] - peaks[0][0]) * 2
    else:  # not enough peaks found
        difference = 0

    plt.plot(peaks[0], np.zeros(len(peaks[0])), "or")
    angularWidth = meterPerPixel * difference / L

    print(str(math.floor(slit) / 10) + "mm : " + str(angularWidth) +
" Rads  or  " + str(difference) + " pixels")

    angleList.append(angularWidth)

    plt.xlabel("Pixel Position")
    plt.ylabel("Pixel Intensity")
    plt.show()
    plt.savefig(str(math.floor(slit) / 10) + ".png")
    plt.cla()
```

```
wavelength = 633e-9
plt.plot(slitRange / 10, angleList, label="Real Data")

expAngles = 2 * wavelength / (slitRange * 1e-4)

plt.plot(slitRange / 10, expAngles, label="Expected")
plt.xlabel("Slit Width (mm)")
plt.ylabel("Angular Width (Rads)")
plt.legend()
plt.show()
```

Appendix B: Heisenberg Uncertainty with Error Propagation

Extends Appendix A to compute the momentum uncertainty from the measured angular widths and plots Δp against 1/Δx. Includes error propagation for uncertainties in the camera distance and peak separation.

```
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import math
from scipy.signal import find_peaks
from scipy.ndimage import gaussian_filter1d

slitRange = np.arange(2, 9, 1)
x = np.arange(0, 1280, 1)
colAvg = 8
meterPerPixel = 5.2e-6  # size of pixel in metres
L = 0.9                  # length from grating to camera
angleList = []
diffList = []
h = 6.63e-34for slit in slitRange:


    im = Image.open(str(math.floor(slit) / 10) +
'mm.png').convert('L')

    na = np.array(im)

    midRowsAvg = np.average(
        na[((np.shape(na)[0] // 2) - colAvg // 2):((np.shape(na)[0]
// 2) + colAvg // 2), :],
```

```
        axis=0
    )
    rowsSmooth = gaussian_filter1d(midRowsAvg, sigma=3)
    fig1 = plt.plot(x, midRowsAvg)

    peaks = find_peaks(rowsSmooth, width=10, distance=150, height=4)

    centralMaxima = np.argmax(rowsSmooth[(peaks[0])])
    if len(peaks[0]) > 2:  # filter out 2nd-order and higher maxima
to get the central two
        difference = (peaks[0][centralMaxima + 1]) -
(peaks[0][centralMaxima - 1])
    elif len(peaks[0]) == 2:  # for images where both outer maxima
couldn't fit; use centre and one side instead
        difference = (peaks[0][1] - peaks[0][0]) * 2
    else:  # not enough peaks found
        difference = 0

    plt.plot(peaks[0], np.zeros(len(peaks[0])), "or")
    angularWidth = meterPerPixel * difference / L

    print(str(math.floor(slit) / 10) + "mm : " + str(angularWidth) +
" Rads  or  " + str(difference) + " pixels")

    angleList.append(angularWidth)
    diffList.append(difference)

    plt.xlabel("Pixel Position")
    plt.ylabel("Pixel Intensity")
    plt.show()
    plt.savefig(str(math.floor(slit) / 10) + ".png")
    plt.cla()  # clean for next run

wavelength = 633e-9  # HeNe laser wavelength in metres
plt.plot(slitRange / 10, angleList, label="Real Data")

expAngles = 2 * wavelength / (slitRange * 1e-4)  # 2*lambda / a
(slit width)

plt.plot(slitRange / 10, expAngles, label="Expected")
plt.xlabel("Slit Width (mm)")
plt.ylabel("Angular Width (Rads)")
```

```
plt.legend()
plt.show()

deltaP = np.array(angleList) * (h / wavelength)  # equation for
momentum uncertainty
deltaXreciprocal = 1 / (slitRange * 1e-4)
plt.cla()
plt.plot(deltaXreciprocal, deltaP)

# Uncertainties
sigmaL = 0.001    # in metres
sigmaDiff = 50    # in pixels

sigmaP = np.array(deltaP) * np.sqrt(
    (np.array(sigmaDiff) / diffList) ** 2 + (sigmaL / L) ** 2
)  # error propagation equation
```

Appendix C: Sinc Function Fitting to Diffraction Pattern

Loads a filtered diffraction image, overlays a theoretical sinc^2 envelope onto the measured intensity profile, and identifies peak positions to compute the angular width.

```
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import math
from scipy.signal import find_peaks
from scipy.ndimage import gaussian_filter1d

slitRange = [20]
x = np.arange(0, 1280, 1)
colAvg = 8
meterPerPixel = 5.2e-6  # size of pixel in metres
L = 0.9                  # length from grating to camera
angleList = []
diffList = []
h = 6.63e-34
angWidPerPix = meterPerPixel / L * 5


def sinc(width):
    y = np.sinc((x - 642) / width)
    return y
```

```
for slit in slitRange:
    im = Image.open(str(math.floor(slit) / 10) + '
filter.png').convert('L')

    na = np.array(im)

    midRowsAvg = np.average(
        na[((np.shape(na)[0] // 2) - colAvg // 2):((np.shape(na)[0]
// 2) + colAvg // 2), :],
        axis=0
    )
    rowsSmooth = gaussian_filter1d(midRowsAvg, sigma=3)
    fig1 = plt.plot(x, midRowsAvg, label="Diffraction Grating")
    print(midRowsAvg[600:700])

    plt.plot(x, (12.25 * sinc(245)) ** 2, label="Single Slit")

    peaks = find_peaks(rowsSmooth, width=10, distance=100, height=4)

    centralMaxima = np.argmax(rowsSmooth[(peaks[0])])
    if len(peaks[0]) > 2:  # filter out 2nd-order and higher maxima
to get the central two
        difference = (peaks[0][centralMaxima + 1]) -
(peaks[0][centralMaxima - 1])
    elif len(peaks[0]) == 2:  # for images where both outer maxima
couldn't fit; use centre and one side instead
        difference = (peaks[0][1] - peaks[0][0]) * 2
    else:  # not enough peaks found
        difference = 0

    plt.plot(peaks[0], np.zeros(len(peaks[0])), "or")
    angularWidth = meterPerPixel * difference / L

    print(str(math.floor(slit) / 10) + "mm : " + str(angularWidth) +
" Rads  or  " + str(difference) + " pixels")

    angleList.append(angularWidth)
    diffList.append(difference)

    plt.xlabel("Pixel Position")
    plt.ylabel("Pixel Intensity")
```

```
    plt.legend()
    plt.show()
    plt.savefig(str(math.floor(slit) / 10) + ".png")
    plt.cla()  # clean for next run

wavelength = 633e-9  # HeNe laser wavelength in metres

# Uncertainties
sigmaL = 0.001    # in metres
sigmaDiff = 50    # in pixels

# sigmaP = np.array(deltaP) *
np.sqrt((np.array(sigmaDiff)/diffList)**2 + (sigmaL/L)**2)
```

Appendix D: 2D Carpet Analysis with Vertical and Horizontal Peak Finding

Iterates over a range of camera distances, locates the vertical centre of the diffraction pattern for each image, extracts the corresponding horizontal intensity profile, and concatenates the pixel bands into a 2D "carpet" image for visualisation.

```
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import math
from scipy.signal import find_peaks
from scipy.ndimage import gaussian_filter1d

x = np.arange(0, 1280, 1)
y = np.arange(0, 1024, 1)
colNum = 1024           # number of columns to average over
meterPerPixel = 5.2e-6  # size of pixel in metres
LRange = np.arange(30, 155, 5)  # distance from grating to camera
(mm)
angleList = []
diffList = []
h = 6.63e-34
colAvg = 8

verticalMidpoints = []
horizontalMidpoints = []
```

```
carpet = np.zeros([8, 1280])

# Find the vertical peak to determine the horizontal level
for L in LRange:
    im = Image.open(str(L) + 'mm.png').convert('L')

    na = np.array(im)

    colAvg = np.average(
        na[((np.shape(na)[0] // 2) - colNum // 2):((np.shape(na)[0]
// 2) + colNum // 2), :],
        axis=1
    )
    colsSmooth = gaussian_filter1d(colAvg, sigma=3)

    peaks = find_peaks(colsSmooth, width=10, distance=100, height=4)

    centralMaxima = np.argmax(colsSmooth[(peaks[0])])
    if len(peaks[0]) > 2:  # filter out 2nd-order and higher maxima
to get the central two
        difference = (peaks[0][centralMaxima + 1]) -
(peaks[0][centralMaxima - 1])
    elif len(peaks[0]) == 2:  # for images where both outer maxima
couldn't fit; use centre and one side instead
        difference = (peaks[0][1] - peaks[0][0]) * 2
    else:  # not enough peaks found
        difference = 0

    angularWidth = meterPerPixel * difference / L

    angleList.append(angularWidth)
    diffList.append(difference)
    verticalMidpoints.append(peaks[0])

    # Horizontal central peak finding
    PixelBand = na[((int(peaks[0]))) - 4:((int(peaks[0]))) + 4, :]
    midRowsAvg = np.average(PixelBand, axis=0)
    rowsSmooth = gaussian_filter1d(midRowsAvg, sigma=3)
    fig1 = plt.plot(x, midRowsAvg, label="Diffraction Grating")

    carpet = np.concatenate((carpet, PixelBand), axis=0)
```

```
    hpeaks = find_peaks(rowsSmooth, width=10, distance=100,
height=4)

    centralMaxima = np.argmax(rowsSmooth[(hpeaks[0])])
    if len(hpeaks[0]) > 2:  # filter out 2nd-order and higher maxima
to get the central two
        difference = (hpeaks[0][centralMaxima + 1]) -
(hpeaks[0][centralMaxima - 1])
    elif len(hpeaks[0]) == 2:  # for images where both outer maxima
couldn't fit; use centre and one side instead
        difference = (hpeaks[0][1] - hpeaks[0][0]) * 2
    else:  # not enough peaks found
        difference = 0

    horizontalMidpoints.append(hpeaks[0][centralMaxima])
    plt.plot(hpeaks[0], np.zeros(len(hpeaks[0])), "or")

    plt.xlabel("Pixel Position")
    plt.ylabel("Pixel Intensity")
    plt.legend()
    plt.show()
    plt.savefig(str(L) + ".png")
    plt.cla()  # clean for next run

plt.imshow(carpet, cmap='grey')

wavelength = 633e-9  # HeNe laser wavelength in metres
```